

Abdul Khurram

+1 (236) 865-4820 | arkb25@student.ubc.ca | [linkedin.com/in/abdulkhurram](https://www.linkedin.com/in/abdulkhurram) | github.com/arkb75

EDUCATION & AWARDS

University of British Columbia

Vancouver, BC, Canada

Bachelor of Science in Computer Science (Expected Graduation: May 2027)

Euclid Mathematics Contest, University of Waterloo, Top 5% Internationally

2021

WORK EXPERIENCE

Shopify

Sept. 2026 – Present

Software Engineer Intern

Toronto, ON, Canada

- Built a **Codex developer-tool integration** used across several teams, with Nix-packaged hooks, 9 reusable skills, and sandbox-validated install and upgrade paths.
- Implemented an approved cleanup removing **6,941 obsolete lines across 373 files**, preserving independent feature gates and settings behavior; merge pending.

RBC Royal Bank | *Python, Scala, Kafka, Airflow*

Jan. – Aug. 2026

Software Developer Co-op

Vancouver, BC, Canada

- Engineered **Python/Scala data pipelines** ingesting **50M+ daily events** via Kafka and Airflow, enforcing RBAC/encryption on **petabyte-scale Delta Lake**.
- Optimized **Spark** jobs to cut **test runtime 40%**; rolled back the change after production instability.
- Worked on multi-region services on **Kubernetes**-orchestrated clusters with fault tolerance, idempotent re-runs, and **Terraform**-provisioned failover.

Contello.ai | *Python, TypeScript/Node.js, AWS, Docker*

Jan. – Dec. 2025

Software Engineer Intern

Toronto, ON, Canada

- Built async **Python** data pipelines on **AWS** (SQS, S3, VPC) processing **thousands of jobs/day** with dead-letter routing and back-pressure controls.
- Raised service reliability to **99.9%** via retry/backoff, Elasticsearch-indexed logs, CloudWatch alerting, and structured observability across distributed services.

RESEARCH EXPERIENCE

UBC Emerging Media Lab | *C++, Python, Unreal Engine*

Sept. – Dec. 2024

Software Developer & Team Lead

Vancouver, BC, Canada

- Led development of a real-time VR simulation in **C++**, designing a data collection pipeline streaming sensor events for AI decision-making.
- Built a multithreaded supervisory system (**C++/Python**) coordinating concurrent actors with shared state and lock-free message passing.
- Profiled the processing pipeline, identified memory and I/O hotspots, and restructured the call graph, cutting latency by **40%** and costs by **90%**.

PROJECTS

Backer | *Go, Python, AWS (DynamoDB, S3), XGBoost*

Hack The Coast 2026

- Designed a **DynamoDB**-backed data layer (10+ tables) with batch operations, transactional writes, and typed abstractions for high-throughput workflows.
- Built a **Go** data ingestion service consuming **DynamoDB** event streams; engineered features and trained an XGBoost ranking model scoring entities on 6 signals.

Workbase | *TypeScript, PostgreSQL, AWS Bedrock, Vitest*

Personal Project

- Built a career content platform that ingests GitHub repos via OAuth, normalizes commit/PR data into **PostgreSQL**, and generates verified highlights with evidence attribution.
- Designed a multi-step **data pipeline** with LLM generation, schema validation, output repair, audit logging, deterministic retry, and automated tests.

SKILLS

Languages: C/C++, Python, Go, Scala, Java, SQL, TypeScript/JavaScript

Infrastructure: Kubernetes, Docker, Terraform, Kafka, Airflow, Apache Spark, AWS (S3, SQS, Lambda, DynamoDB, CloudWatch); Linux, Git, CI/CD

Databases & Storage: PostgreSQL, DynamoDB, Elasticsearch, Redis, Delta Lake; REST APIs, Flask/FastAPI